

Well I tried following your steps and my calculus started coming back. I understood, so for the Bias, would be the same but DLoss/DB?



Exactly! You've got the main idea.

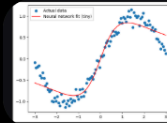
I looked through your work, and your derivation for the weight is essentially correct. You:

- Started with the MSE.
- Substituted $\hat{y} = wx + b$.
- Applied the chain rule.
- Recognized that $\frac{d}{dw}(wx + b - y) = x$.
- Arrived at

$$\frac{\partial L}{\partial w} = \frac{1}{n} \sum_{i=1}^n 2(\hat{y}_i - y_i)x_i.$$

That's exactly what the implementation uses.

Now for the bias, you follow the exact same process except every instance of $\frac{d}{dw}$ becomes $\frac{d}{db}$.



I'm glad!
It means that for plotting, it becomes a single vector?
Also, does it look correct here, in the image?

Yes! That's exactly what it becomes.

Think of it this way.

During training

The network wants a **data matrix**:

`</> Python`

```
x.shape == (100, 1)
```



Run

which means

```
[ -3.00  
  -2.94  
  -2.88  
   ...  
   3.00
```



100 rows x 1 column

Each row is one sample, and the one column is the le input feature.

i have a tensor mismatch
issue now

Worked for 18s >

The mismatch comes from using stale
`logits` in the training evaluation loop.

Your final optimization batch contains 18
samples, so `logits` has shape:

```
python  
[18, 7]
```

The current evaluation batch contains 32
labels, so `yb` has shape:

```
python  
[32]
```

At line 1175, you use:

```
python  
predicted = logits.argmax(dim=1)
```

But the current evaluation predictions are
stored in `outputs`, not `logits`.

Correct that training evaluation loop to:

```
python  
for xb, yb in experiment_train_
```

Ask for follow-up changes



5.6 Sol



2.12)

I don't want you to make the changes, I just want guidance to fix issues that arise, especially for sanity checks.

Now with that being said, I need you to check if my wrapped function for my training loop with torch is right

Worked for 1m 1s >

Your wrapped function is not correct yet. I didn't modify the notebook.

The main problems are:

1. The training loop is outside the epoch loop

At [Lab_3.ipynb \(line 1168\)](#), only these operations are inside `for epoch`
`in range(epochs) :`

```
python
model.train()
train_loss, train_correct, trai
```

The batch loop at line 1173 is dedented. Consequently, the function loops over `epochs` doing nothing useful, then trains for exactly one epoch and returns one history entry.

...

Everything from:

2.12)

```
python
```

```
for xb, yb in train_loader:
```

through history appending must be inside the epoch loop.

2. The experiment parameters are mostly ignored

At [Lab_3.ipynb \(line 1157\)](#), the function accepts:

```
python
```

```
hidden_sizes, lr, batch_size, e
```

But currently:

- `hidden_sizes` is never used.
- `dropout` is never used.
- `batch_size` is never used.
- `lr` is ignored because line 1160 hardcodes `lr=1e-3`.
- Only `epochs` affects execution—and because of indentation, it still does not control actual training epochs correctly.

The optimizer should use:

```
python
```

```
optimizer = torch.optim.Adam(mo
```

The training loader must be constructed inside the function `def train`

2.12)

```
python

model = FeedForwardNet(
    input_dim=X_train_scaled.sh
    hidden_sizes=hidden_sizes,
    dropout=dropout,
).to(DEVICE)
```

You will first need to update `FeedForwardNet` to accept `dropout`.

4. The current network cannot support the requested architectures

Your model directly accesses `hidden_sizes[0]` and `hidden_sizes[1]` at [Lab_3.ipynb \(line 873\)](#).

Therefore:

- `hidden_sizes=(8,)` will raise an `IndexError`.
- `hidden_sizes=(256, 128, 64)` will silently ignore `64`.
- There is nowhere to insert the requested dropout.

The model needs to build its layers dynamically by iterating through `hidden_sizes`, typically using a list of `nn.Linear`, `nn.ReLU`, and optional `nn.Dropout` layers inside `nn.Sequential`.

5. Validation and test data are not scaled

This is likely the cause of your extremely large validation loss.